

CHAPTER 10

Actuators

10.1. Space story

The new Series 5000 satellites had a PID controller for station keeping. On the first station-keeping maneuver, we noticed that it was oscillating. The problem was that there was a high-gain instability, which can happen with a PID because, at the minimum pulsewidth, where the valve is opened for the shortest possible time, the thruster produced more impulse than expected. This was fixed by adjusting the PID gains. This was the first time we had noticed this problem.

10.2. Introduction

The performance of a spacecraft-control system is limited by the performance of its sensors and actuators. This chapter discusses the types of actuators that are used in spacecraft-control systems. Many of the actuators are used just for attitude control. Others are used for orbit adjustment and station keeping, as well as attitude control.

10.3. Types of actuators

Table 10.1 lists some actuators used in spacecraft. We refer to attitude control and momentum unloading. Systems that rely on momentum exchange (reaction wheels, momentum wheels, and control-moment gyros) need momentum unloading, that is an inertial torque, to compensate for inertially fixed-attitude disturbances.

Table 10.1 Classes of Actuators.

Type	Description	Application
Magnetic torquer air coil	Wire wrapped around the frame of the spacecraft to form a planar loop.	Momentum unloading, two-axis control. Average three-axis control.
Magnetic-torquer bar	Wire wrapped around a magnetic steel core, often multiple coils around a single core. Iron core torquers are usually operated in saturation. The second coil can produce 20% more torque, which is useful in an emergency.	Momentum unloading, two-axis control. Average three-axis control.

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Table 10.1 (*continued*)

Type	Description	Application
Thruster hydrazine	Monopropellant hydrazine is passed through a catalyst bed that causes an exothermic reaction. This produces thrust.	Attitude and orbit control.
Thruster bipropellant	An oxidizer and fuel are mixed to produce a chemical reaction.	Attitude and orbit control.
Thruster cold gas	Pressurized gas is expanded through a nozzle.	Attitude Control.
Electrothermal hydrazine thrusters (EHT)	Same as hydrazine except that it has a heating coil to heat the exhaust of the exothermic reaction.	Attitude control.
Thruster arc jet	Same as hydrazine except that it has an electric arc to heat the exhaust of the exothermic reaction.	Orbit changes.
Pulsed plasma thruster (PPT)	Ionized gas is accelerated in a magnetic field.	Attitude and orbit control.
Hall thruster	Thruster uses the Hall effect to accelerate ions.	Orbit control.
Ion engine	Thruster in which ions are accelerated using a charged grid.	Orbit control.
Magnetoplasmadynamic Thruster (MPD)	Magnetic field accelerates ions. Lithium is the only practical fuel.	Orbit control.
Variable Specific Impulse Magnetoplasma Rocket (VASIMR)	An electric rocket composed of three cells. The first creates a plasma. The second further heats it with radio-frequency waves the third is a magnetic nozzle that converts it to a directional flow separate from the magnetic-field lines.	Orbit control.
Reaction wheel	A wheel is attached to the spacecraft via an electric motor. Provides a means of regulated momentum exchange between a flywheel and the rest of the spacecraft through reaction torques. The motor uses current feedback to linearize the torque response. This compensates for back-electromotive force.	Attitude control.
Momentum wheel	A wheel is attached to the spacecraft via an electric motor. Provides a means of regulated momentum exchange between a flywheel and the rest of the spacecraft through reaction torques. The wheel spins at a high rate to provide a momentum bias. It is modulated around this speed to control the attitude along the spin axis.	Attitude control.

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Table 10.1 (continued)

Type	Description	Application
Pivoted wheel	Reaction wheel on a single or two-degree-of-freedom platform. A pivoted wheel has a limited range of angular motion, in contrast with a control-moment gyro that has large-angle rotation capability. In a pivoted wheel both the rotation of the wheel about its axis and the rotation of the wheel assembly is used for control.	Attitude control.
Control-moment gyro	A spinning wheel is attached to the spacecraft through one or two single-degree-of-freedom hinges. The spin rate is maintained and torques are generated normally only about the pivot axes. Some systems also vary the wheel speed for an additional control degree-of-freedom.	Attitude control of slew maneuvers.
Solar paddles	Solar-pressure torques are generated by moving the paddles. This is normally done with solar panels or with solar sails.	Attitude control and momentum unloading.

10.4. Reaction-wheel model

10.4.1 Introduction

Reaction wheels are momentum-exchange devices. A motor is fixed to the spacecraft and the shaft of the motor is attached to a flywheel. When a control voltage is applied to the motor, the motor generates a torque spinning the wheel in one direction and the spacecraft in the other; hence the term reaction wheel. Since the torque is internal to the spacecraft system, the reaction wheel cannot change the total inertial angular momentum of the spacecraft system. Rather, it transfers momentum between the flywheel and the rest of the spacecraft. If the external torque on the spacecraft is periodic (with a sufficiently high frequency) with respect to the inertial frame, then the reaction wheel can completely control the spacecraft. However, if there is a constant inertial torque, the wheel will spin up and eventually saturate. The wheel can also be used to maneuver the spacecraft. When the wheel spins up the spacecraft will develop an angular rate. When the wheel spins down it will absorb the angular momentum in the body and stop the motion. See Fig. 10.1.

Fig. 10.2 shows the anatomy of an axial-flux reaction wheel. It uses a three-phase stator and a Halbach array rotor. The angle of the rotor is measured from the currents in the stator and the measured angle from the angle encoder. Axial flux means that the magnet flux is along the spin axis. The Halbach magnet configuration concentrates the flux on one side of the magnet, increasing the available torque.

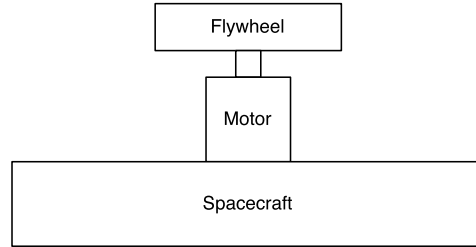


Figure 10.1 Reaction wheel.



Figure 10.2 Anatomy of an axial-flux motor reaction wheel. Clockwise from left: side view showing the stator coils, the Halbach rotor magnet array, the complete wheel in operation, view from the top and bottom view showing the optical angle encoder.

10.4.2 Momentum exchange

The single-axis reaction wheel and spacecraft model is

$$H = I\omega + J(\Omega + \omega) \quad (10.1)$$

$$T = \dot{H} = I\dot{\omega} + J(\dot{\Omega} + \dot{\omega}) \quad (10.2)$$

$$T_w = J(\dot{\Omega} + \dot{\omega}) \quad (10.3)$$

where H is the inertial angular momentum, T is the body torque, I is the body inertia, J is the wheel inertia, Ω is the wheel angular rate, and ω is the body angular rate.

If the attitude control system is perfect then

$$\dot{\omega} = 0 \quad (10.4)$$

$$T = J\dot{\Omega} \quad (10.5)$$

The wheel absorbs all of the momentum from the external torque.

10.4.3 Motor model

Most reaction wheels are driven by DC motors. A simplified model is shown in Fig. 10.3. The incoming torque command is converted into a current by the first block.

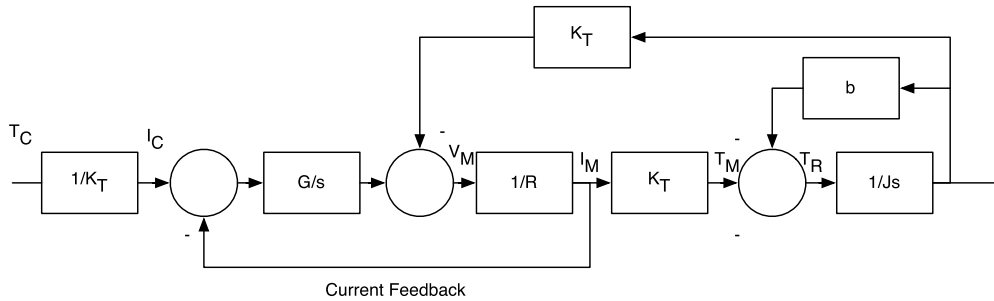


Figure 10.3 DC motor model.

The summing junction subtracts the motor current. This current feedback loop compensates for the back-electromotive force that is a function of speed. This permits the device to be used as a torque actuator.

The difference between the commanded current and feedback current is integrated and multiplied by a gain. The back-electromotive force (back emf) voltage is subtracted, resulting in the motor voltage. The motor voltage is limited by the spacecraft bus voltage. The motor voltage is divided by the motor resistance to get the motor current, which is then multiplied by the motor gain to produce the motor torque. The frictional torques are subtracted from this to get the reaction torque. When using the wheel as a torque actuator, the transfer function from commanded torque to reaction torque is of interest. The current loop compensates for the back-emf. To compensate for the friction torques, another loop would be needed that fed back rate. This is often done in robotics where precise torque control is necessary.

The transfer function from commanded torque to reaction torque is

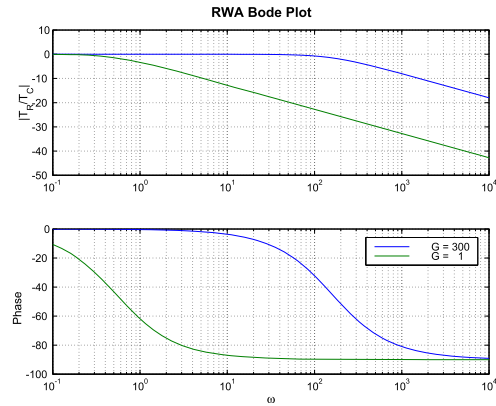
$$\frac{T_R}{T_C} = \frac{G}{R} \frac{s}{s^2 + \left(\frac{b}{J} + \frac{k_T^2}{RJ} + \frac{G}{R} \right) s + \frac{bG}{JR}} \quad (10.6)$$

where J is the reaction-wheel inertia, b is the viscous damping coefficient, R is the motor resistance, k_T is the current to motor torque gain, and G is the forward gain. There is no DC response since when a constant command is supplied the wheel spins up and the viscous friction torque eventually cancels the motor torque. At high frequencies, the RWA gain rolls off as $1/s$. At intermediate frequencies, if G is large, the gain is nearly 1.

```

1 i      = 0.0398; kT = 0.0329; r = 1.8856; b =
   0.001;
2 g      = 300;
3 num     = [g/r 0];
4 den     = [1 b/1i+kT^2/(r*1i) + g/r b*g/(r*1i)];
5 w      = logspace(-1,4);
6 [m1, p1] = FResp( num, den, w );
7 g      = 1;
8 num     = [g/r 0];
9 den     = [1 b/1i+kT^2/(r*1i) + g/r b*g/(r*1i)];
10 [m2, p2] = FResp( num, den, w );
11 m1     = 10*log10(m1); m2 = 10*log10(m2);
12 h      = Plot2D( w, [m1;m2;p1;p2], '\omega', ...
13               {'|T_R/T_C|' 'Phase'}, 'RWA_Bode_Plot'
14               '\xlog', {'[1 2]' '[3 4]'} );
15 legend('G_u=300', 'G_u=1');

```



Example 10.1: Reaction-wheel frequency response.

10.4.4 Reaction-wheel state equations with current feedback

There are two states in this model, the wheel speed (a mechanical state) and the integral of the current difference (an electrical state). The state equations are derived as follows. The reaction torque is

$$T_R = k_T i_M - b\omega - T_F \quad (10.7)$$

where T_F is all nonviscous friction torques. The motor current is

$$i_M = \frac{1}{R}(V_M - k_T \omega) \quad (10.8)$$

The motor torque is

$$V_M = G \int (i_M - i_C) \quad (10.9)$$

The electrical state derivative is

$$\dot{z} = i_M - i_C \quad (10.10)$$

The mechanical state derivative is

$$\dot{\omega} = \frac{T_R}{J} \quad (10.11)$$

The state equations are

$$\begin{bmatrix} \dot{\omega} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \frac{k_T^2}{RJ} & \frac{bk_T G}{J^2 R} \\ \frac{k_T}{R} & \frac{G}{R} \end{bmatrix} \begin{bmatrix} \omega \\ z \end{bmatrix} + \begin{bmatrix} 0 & 1/J \\ 1/k_T & 0 \end{bmatrix} \begin{bmatrix} T_C \\ T_F \end{bmatrix} \quad (10.12)$$

The electrical dynamics are significant at much higher frequencies than the mechanical dynamics. They can be removed from the equations by setting the derivative of z equal to zero and finding the steady-state value of z , which is

$$z = \frac{G}{R} \left(\frac{T_C}{k_T} + \frac{k_T}{R} \omega \right) \quad (10.13)$$

The reaction-wheel actuator can then be modeled with a single state equation. This is the equivalent of saying that $i_M = i_C$. The state equations are linear and do not include voltage and current limits. When implemented in code, this must be included. Fig. 10.4 shows a doublet (a positive then a negative torque) with and without current feedback. The effect of back-emf is evident.

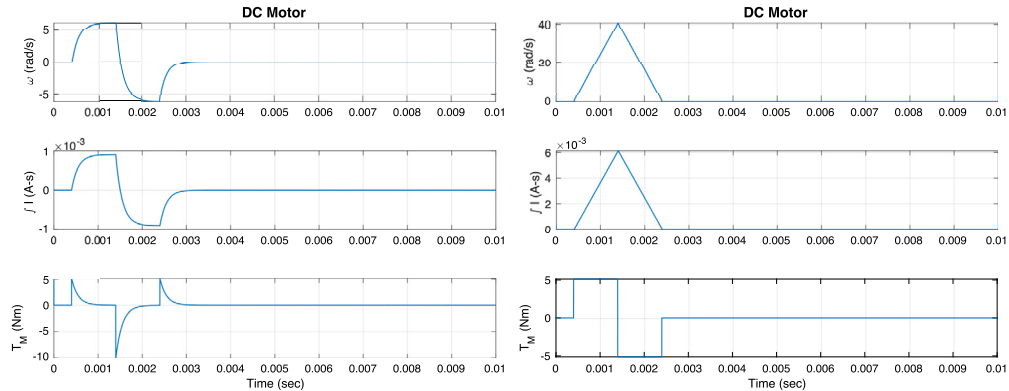


Figure 10.4 Response to a doublet with and without current feedback. The plot on the right shows current feedback.

10.4.5 Tachometer

Most spacecraft use permanent magnet motors. The rotor has poles made up of rare-Earth magnets. The stator currents are switched based on the angle of the magnets with respect to the stator. The angle is determined by Hall-effect sensors that give a pulse when the magnet pole passes over the switch. The time between Hall-sensor pulses can be used to determine the wheel rate. Motors can have many poles; For a 3-phase motor, only 3 Hall sensors are needed for commutation. However, up to 18 sensors are not unusual. Time is computed by running a counter and dumping the count when a

Hall sensor signals. If the motor is running fast, one count per cycle is sufficient. If it is running slowly, the count between each Hall sensor is output. If the motor is running sufficiently slow, there may not be an update available between samples. This presents problems for low-speed operation. One solution is to use current measurements to estimate the rate at low speed. Another is to put an angle encoder on the shaft.

10.4.6 Friction

Friction is an important consideration when modeling reaction wheels or any rotating or sliding component. The simplest model is that of Coulomb friction and viscous friction.

$$F = -\frac{\nu}{|\nu|}f_c - \sigma_\nu \nu \quad (10.14)$$

The first term is a constant force (or torque) that resists motion and is independent of speed. The second term is proportional to the speed (or angular rate). This model does not include stiction, also known as breakaway friction, which is the force (or torque) needed to get something moving. This can be modeled using the bristle-friction model [1]. This model adds a state that can be thought of as the bending of the bristles. The model is

$$F = \sigma_0 z + \sigma_1 \dot{z} + \sigma_\nu * \nu \quad (10.15)$$

where

$$\dot{z} = \nu - \frac{|\nu|z}{g(\nu)} \quad (10.16)$$

and

$$g(\nu) = \frac{f_C + (f_S - f_C) * e^{-\left(\frac{\nu_S}{\nu}\right)^2}}{\sigma_0} \quad (10.17)$$

Example 10.2 compares bristle friction with Coulomb friction. The parameters in the model are not readily available from vendors and must be obtained from testing.

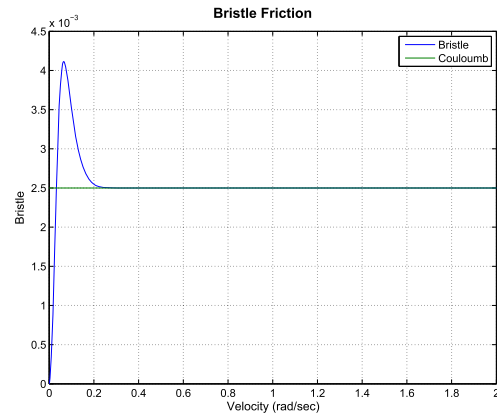
10.4.7 Zero crossings

Near to zero wheel speed the stiction torque tries to keep the wheel at zero speed. With trapezoidal commutation using Hall sensors for rotor-angle measurements it is difficult to measure the wheel speed since you are not getting a steady stream of Hall-sensor pulses. One strategy is to bias up the wheels to avoid zero crossings. This requires more momentum storage but should improve the wheel life.


```

1 dT = 0.0125;
2 nSim = 1000;
3 fCoul = 0.005/2;
4
5 dRHS.fStatic = 0.005;
6 dRHS.fCoulomb = fCoul;
7 dRHS.vStribeck = 0.1;
8 dRHS.sigma0 = 1;
9 dRHS.sigma1 = 1e-4;
10 dRHS.sigma2 = 0;
11 dRHS.maxC = 1/dT;
12
13 xPlot = zeros(2,nSim);
14 x = 0;
15 v = linspace(0,2,nSim);
16
17 for k = 1:nSim
18     dRHS.v = v(k);
19     x = RK4( @FrictionBristle, x, dT,
20             0, dRHS );
21     [zDot, f, c] = FrictionBristle( x, 0, dRHS );
22     xPlot(:,k) = [f;fCoul];
23 end
24 Plot2D( v, xPlot, 'Velocity(rad/sec)', 'Bristle',
25         'BristleFriction','lin')
26 legend({'Bristle' 'Coulomb'});

```



Example 10.2: Bristle friction and Coulomb friction.

10.4.8 Commutation

Commutation for motors with a trapezoidal drive uses Hall sensors (magnetic-field sensors). Six-state commutation directs current through two phases of the motor at a time, based on the Hall-sensor switches. The current switches when a Hall sensor changes state. The torque angle curve is not always at the ideal 90 degrees, resulting in torque ripple.

Table 10.2 shows the commutation pattern.

Table 10.2 Reaction-wheel commutation pattern.

Phase A	Phase B	Phase C
OFF	+	-
-	+	OFF
-	OFF	+
OFF	-	+
+	-	OFF
+	OFF	-

Fig. 10.5 shows the response to commutation. Note the torque noise.

10.4.9 Suspensions

Noise sources due to reaction wheels are:

1. Center-of-mass is not on the spin axis;

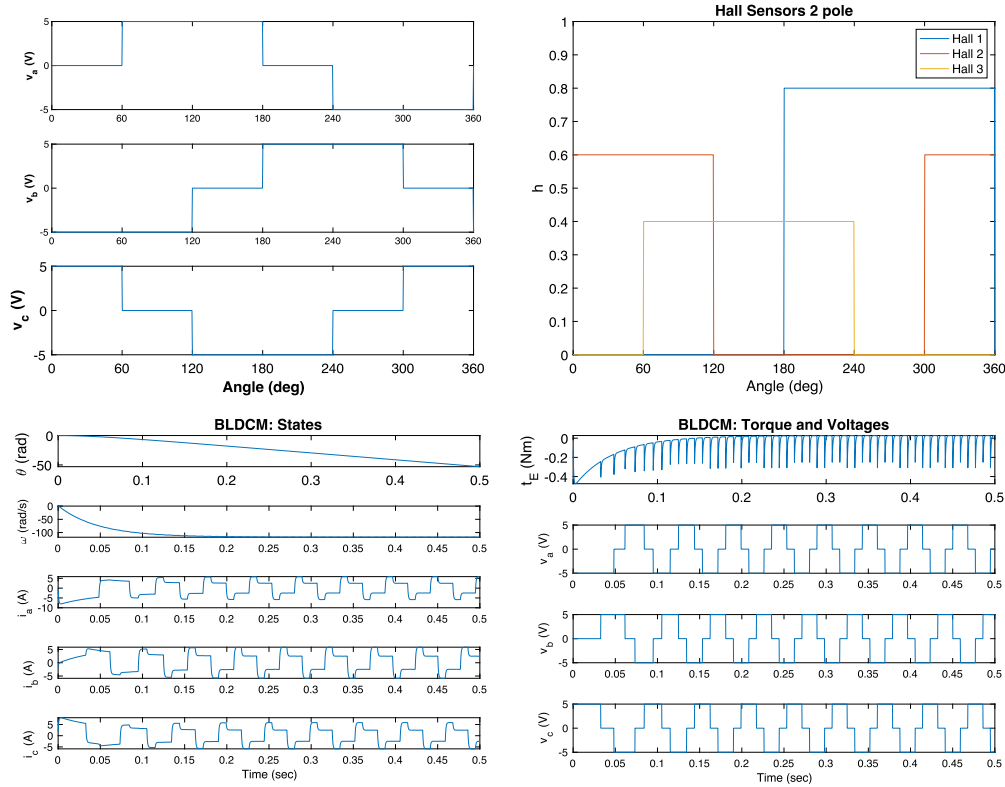


Figure 10.5 Response to commutation.

2. Nondiagonal inertia matrix;
3. Angular acceleration about z produces torque around all three axes;
4. Torque ripple;
5. Torque production is not linear due to commutation;
6. Cogging due to interaction of magnets with the stator steel;
7. Bearing noise.

For precision pointers (James Webb, Hubble) suspensions are used.

Fig. 10.6 shows the suspension for the NASA Hubble Space Telescope wheels. The system uses a spring and a damper.

10.5. Control-moment gyro

10.5.1 Introduction

Reaction wheels are limited in torque capability and momentum-storage capability. When higher torques are needed, for example on a reconnaissance satellite that must

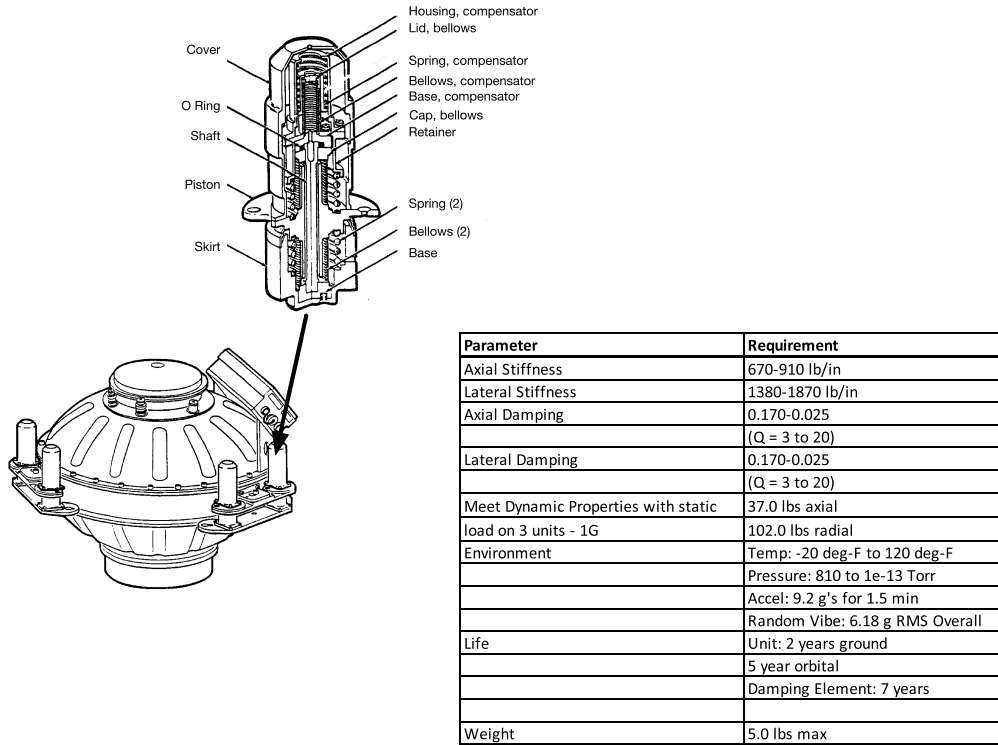


Figure 10.6 Hubble Space Telescope suspension [2]. Image courtesy of NASA.

track a target on the Earth, control-moment gyros (CMGs) are often used. A CMG is a spinning wheel on a gimballed platform. CMGs may have one or two gimbals. The wheel spin rate is kept constant with a tachometer loop. This degree-of-freedom is not used for control. The gimbal torquer motors are used for control.

10.5.2 Modeling

The CMG adds another body to the spacecraft with 2 or more degrees-of-freedom. Generally, the inner gimbal is limited to ± 90 degrees, while the outer gimbal can rotate 360 degrees. Power is supplied to the inner gimbal via slip rings. Any spacecraft with CMGs is a multibody spacecraft. For control-design purposes, however, it is useful to model the CMG drive as a high-bandwidth drive in which the gimbal rates achieve a commanded rate instantaneously. The output torque for one CMG is

$$T = \dot{B}H \quad (10.18)$$

where H is the rotor angular momentum and is constant. B is composed of two parts. One is constant and represents the alignment of the CMG base. The second includes the contributions due to the gimbal rotations. Assume that the rotor, in the CMG base frame, is aligned along the y -axis. Then, the torque equation becomes

$$T = Bh \begin{bmatrix} \cos \alpha & 0 \\ -\cos \beta \sin \alpha & -\sin \beta \cos \alpha \\ -\sin \beta \sin \alpha & \cos \beta \cos \alpha \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \end{bmatrix} \quad (10.19)$$

where B is the transformation matrix from the CMG base to the spacecraft frame. When α and β are zero, the CMG momentum for this particular selection of gimbal angles, in the CMG frame, is

$$H = \begin{bmatrix} 0 \\ h \\ 0 \end{bmatrix} \quad (10.20)$$

10.5.3 Torque distribution

Each double-gimbal CMG has two control degrees-of-freedom. At least two CMGs are need for 3-axis control, but even two will have four degrees-of-freedom. One way to distribute the torque demand is through a pseudoinverse law that penalizes the squared magnitude of the gimbal rates. If

$$T = A \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \end{bmatrix} \quad (10.21)$$

is the torque for one CMG then

$$T = \begin{bmatrix} A_1 & A_2 \end{bmatrix} \begin{bmatrix} \dot{\alpha}_1 \\ \dot{\beta}_1 \\ \dot{\alpha}_2 \\ \dot{\beta}_2 \end{bmatrix} \quad (10.22)$$

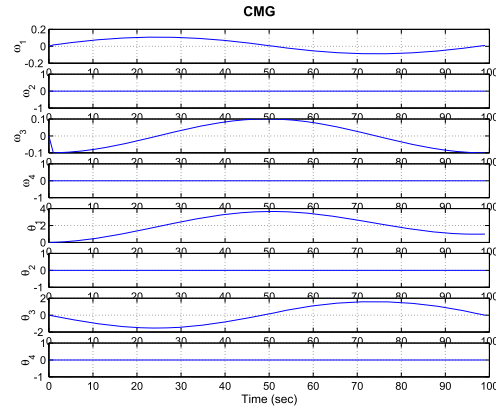
is the torque for two CMGs. The gimbal rates are then

$$\begin{bmatrix} \dot{\alpha}_1 \\ \dot{\beta}_1 \\ \dot{\alpha}_2 \\ \dot{\beta}_2 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 \end{bmatrix}^T \left(\begin{bmatrix} A_1 & A_2 \end{bmatrix}^T \begin{bmatrix} A_1 & A_2 \end{bmatrix} \right)^{-1} T \quad (10.23)$$

This torque distribution law only requires the inverse of a 3-by-3 matrix. However, while it attempts to minimize the gimbal rates it does not try to keep the CMGs away from gimbal lock, a condition in which control is lost along one degree-of-freedom. If large CMG motion is expected, a more sophisticated approach will be required. Example 10.3 shows the gimbal response to an x and y disturbance torque.

```

1 h = 10; n = 100; dT = 1.0;
2 b = {eye(3) [0 1 0;-1 0 0;0 0 1]};
3 d = linspace(0,2*pi,n);
4 t = [sin(d)+0.1;cos(d);zeros(1,n)];
5 omega = zeros(4,n); theta = zeros(4,n);
6 a = zeros(3,4);
7 for k = 2:n
8     i = 1;
9     for j = 1:2:4
10        cA = cos(theta(j,k)); sA = sin(theta(j,k));
11        cB = cos(theta(j+1,k)); sB = sin(theta(j+1,k));
12        a(:,[j j+1]) = b{i}*[cA 0;-sB*sA -sB*cA;-sB*sA cB*cA]*h;
13        i = i + 1;
14    end
15    omega(:,k) = pinv(a)*t(:,k);
16    theta(:,k) = theta(:,k-1) + dT*omega(:,k);
17 end
18 time = (0:(n-1))*dT;
19 Plot2D(time,[omega;theta],'Time(sec)',...
20         {'\omega_1','\omega_2','\omega_3','\omega_4',...
21          '\theta_1','\theta_2','\theta_3','\theta_4'},'CMG');
```



Example 10.3: Double-gimbal CMG.

10.5.4 Single-axis control-moment gyros

A single-axis control-moment gyro has a single-axis hinge between the gyroscope assembly and the spacecraft. This constraint implies that the momentum in the wheel will be seen as a momentum bias on the spacecraft. The control torque on the spacecraft is due to the precession of the wheel assembly.

$$T = B\Omega^\times h \quad (10.24)$$

where h is the wheel-momentum vector in the CMG frame, Ω is the precession rate vector, and B transforms from the base CMG frame to the body frame. The skew-symmetric matrix is

$$\Omega^\times = \begin{bmatrix} 0 & -\Omega_z & \Omega_y \\ \Omega_z & 0 & -\Omega_x \\ -\Omega_y & \Omega_x & 0 \end{bmatrix} \quad (10.25)$$

If we define that in the CMG frame, the nominal momentum vector is along z and the hinge axis is along x we get

$$T = B \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \delta & -\sin \delta \\ 0 & \sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\dot{\delta} \\ 0 & \dot{\delta} & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ h \end{bmatrix} = -B \begin{bmatrix} 0 \\ \cos \delta \\ \sin \delta \end{bmatrix} h \dot{\delta} \quad (10.26)$$

When $\delta = 0$

$$T = -B \begin{bmatrix} 0 \\ h \dot{\delta} \\ 0 \end{bmatrix} \quad (10.27)$$

By picking the appropriate B matrices we can generate 3-axis torques. However, as the gimbal angles change we will find that holes in the momentum envelope may appear.

10.6. Thrusters

10.6.1 Introduction

Small monopropellant and bipropellant thrusters are used for attitude control on many satellites. They are used for attitude control during orbit-change maneuvers or momentum unloading, as an alternative to magnetic torquers. Most small thrusters are on-off thrusters. That is, their valves have only two positions, on and off.

10.6.2 Pulsewidth modulation

An alternative is to control the pulse length of the thruster. If the control period is 1 second, for example, the average thrust can be halved if the pulsewidth is 0.5 s. The thrust does not change, but the average thrust over the period of control is halved. This is done at the expense of adding high-frequency control activity. For example, if a constant thrust of 1/2 the thruster thrust is desired, the actual thrust will be a square wave with a 50% duty cycle and a cycle period equal to the control period. Thus, the actual output of the thruster will have significant high-frequency harmonics and might excite vibratory modes in the system.

10.6.3 Minimum impulse bit

A major problem is that due to the dynamics of the valve, all thrusters have minimum impulse bits. Generally, 16 ms is the minimum on-time for a thruster. If the control period is 1 second, that gives an effective throttle ratio of 62.5:1. That is the equivalent of a 6-bit data word! In contrast, an audio CD uses 16-bit data words! This can be improved by using a multirate control system. The digital controller runs with a period

of 1 second, but the output is sampled less frequently. For example, the output might be sampled every 8 control periods. This improves the control resolution to 9 bits.

When designing a control system it is important to include the effects of under-sampling. The easiest way to do this is to add a zero-order hold with a duration of the minimum impulse bit, followed by a delay of length equal to the period between output samples.

Fig. 10.7 shows how the impulse bit can vary with pulsewidth for a monopropellant thruster. The spike at the minimum pulsewidth increases the loop gain and can cause

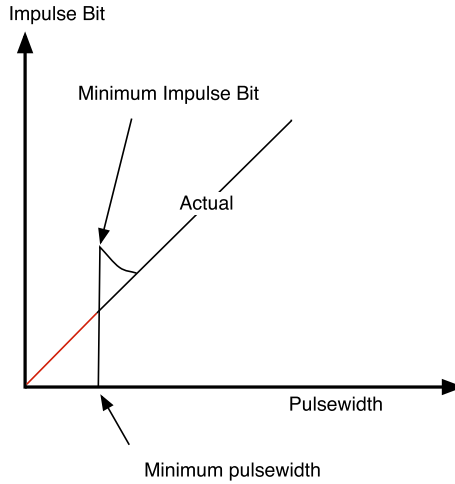


Figure 10.7 Impulse bit vs. pulsewidth.

problems with proportional-integral-derivative (PID) controllers as they are conditionally stable.

10.6.4 Time constants

Thrusters with solenoid valves can be modeled as first-order systems with separate rise and fall time constants. When fuel is flowing into the thruster the model is

$$\dot{u} + \frac{u}{\tau_f} = 1 \quad (10.28)$$

when no fuel is flowing the model is

$$\dot{u} + \frac{u}{\tau_f} = 0 \quad (10.29)$$

The thrust is

$$T = ku \quad (10.30)$$

where k is the maximum thrust given by

$$k = C_F A^* P \quad (10.31)$$

where P is the pressure in the rocket combustion chamber, C_F is the thrust coefficient and A^* is the throat area.

10.6.5 Fuel system

A typical fuel system is shown in Fig. 10.8. The system has two fuel volumes (that may consist of several tanks) with hydrazine fuel pressurized by helium. The systems

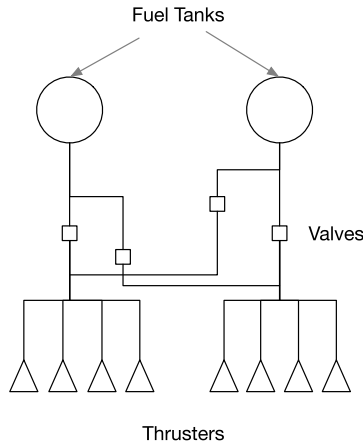


Figure 10.8 Tank geometry.

are cross-strapped (either set of thrusters can use either tank) with a total of four latch valves. Each half system is attached to six REAs.

If the system is unregulated, the pressure in each system is

$$P = \frac{m_{\text{He}} R_{\text{He}} T}{V - \frac{m_{\text{Hydrazine}}}{\rho_{\text{Hydrazine}}}} \quad (10.32)$$

where m_{He} is the mass of helium, R_{He} is the gas constant of helium,

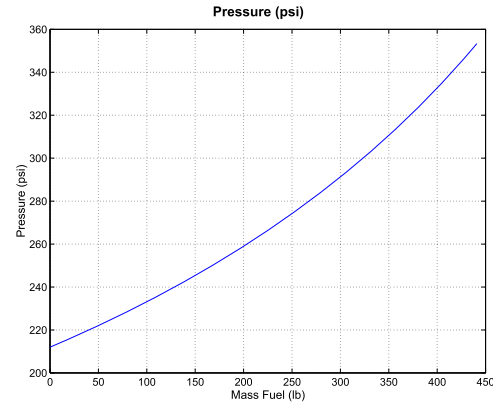
$$R_{\text{He}} = \frac{8.3}{0.004} \quad (10.33)$$

T is the gas temperature, V is the system volume (including any auxiliary pressurant tanks), $m_{\text{Hydrazine}}$ is the mass of the fuel, and $\rho_{\text{Hydrazine}}$ is the density of the fuel. The Demo in Example 10.4 shows a propellant system designed so that the pressure stays between 210 and 250 psi during the life of the mission.


```

1 Nm2ToPSI = 1.4504e-04;
2 KgToLbF = 2.2046e+00;
3 molWt = 0.004;
4 rPress = 2.0786e+03;
5 mFuel = 0.200;
6 rhoFuel = 1000;
7 T = 293;
8 vTank = 0.5;
9 mPress = 1.2;
10 p = BloDown(mPress, rhoFuel, vTank, rPress, T,
              mFuel);
11
12 Plot2D(mFuel*KgToLbF, p*Nm2ToPSI, 'Mass_Fuel(lb)',
         'Pressure_(psi)');

```



Example 10.4: Blowdown curve.

10.7. Magnetic torquers

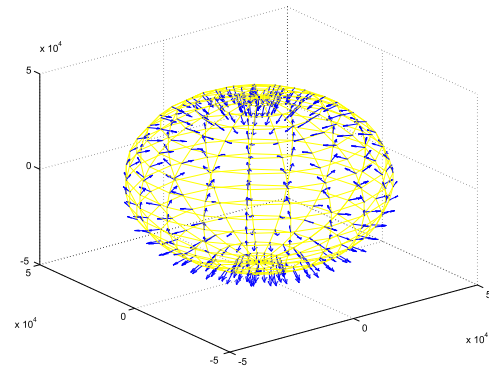
10.7.1 The magnetic field

The magnetic-field vectors are shown in Example 10.5. This particular model represents the field as a tilted dipole, which is good for many low-Earth orbit applications.

```

1 % For a geo orbit
2 DrawFieldLines( 43200, 20, 'BDipole', 2451545.0 )

```



Example 10.5: Magnetic field at the geosynchronous-orbit altitude.

10.7.2 Magnetic torque

10.7.2.1 Torque production

Magnetic torquers are configured as either an air coil or a torquer bar. Torquers generate torque through interaction with the Earth's magnetic field

$$T = M \times B \quad (10.34)$$

where B is the magnetic field and M is the dipole of the torquer. For an air coil, the dipole is

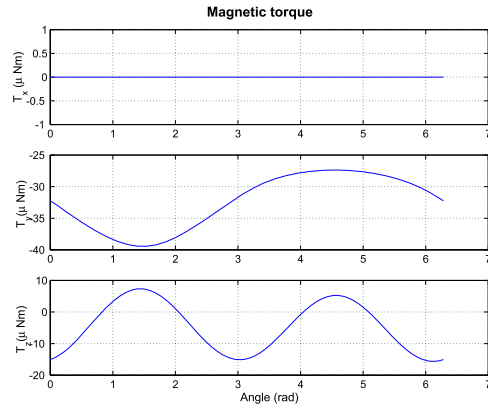
$$M = NIA \quad (10.35)$$

where N is the number of turns of wire, I is the current in the wire, and A is the area enclosed by the coil. Typically, the wire is wrapped around the corners of the spacecraft structure.

Example 10.6 shows the force vector generated by a 400 ATM² (Amp-turn-meters squared) magnetic torquer in any Earth-pointing spacecraft at geosynchronous orbit. The torquer is aligned with the x -axis, which is in the orbit plane. In geosynchronous orbit, the magnetic field is nearly normal to the orbit plane. In this example, the Sun is along the x -axis. The model accounts for solar activity, which is set to a maximum. At geosynchronous orbit, the magnetic field is strongly influenced by the Sun. The torque is in the body frame.

```

1 a = linspace(0,2*pi);
2 c = cos(a);
3 s = sin(a);
4 n = 100;
5
6 % Circular orbit
7 r = 43200*[s;c;zeros(1,n)];
8 m = [400;0;0];
9 jD = Date2JD([2010 3 21 0 0 0]);
10 t = zeros(3,n);
11 for k = 1:n
12     b = [c(k) s(k) 0;-s(k) c(k) 0;0 0 1];
13     t(:,k) = cross(m, b*BMF(r(:,k), jD, 3));
14 end
15 Plot2D(a,t*1e6,'Angle_(rad)',...
16 {'T_x_(\mu Nm)' 'T_y_(\mu Nm)' 'T_z_(\mu Nm)'} ,...
17 'Magnetic_Torque')
```



Example 10.6: Magnetic torque at geo.

Magnetic torquers cannot instantaneously produce a 3-axis torque. One approach is to compute the moments for the dipoles that fit the desired torque in a least-squares sense. Assume that there are n dipoles so that the dipole vector m is $n \times 1$. The dipole unit vector's matrix is u and is $3 \times n$. If the cost of using the torquers is the scalar c and the magnetic-field vector is b define

$$\gamma = u \times b \quad (10.36)$$

The scalar cost is

$$J = (\gamma m - t_D)^T (\gamma m - t_D) + c m^T m \quad (10.37)$$

Taking the partial derivative with respect to m we get the least-square dipole, which is

$$m = (\gamma^T \gamma + \epsilon)^{-1} \gamma t_D \quad (10.38)$$

where t_D is the torque demand.

10.7.2.2 Magnetic-torquer design

Torquer bars use magnetic steel to concentrate the flux, getting the same dipole in a smaller volume. However, the magnetic steel can saturate, limiting the dipole. In addition, the magnetic steel will always have some small residual dipole when off. The power consumed by a torquer is

$$P = \frac{V^2}{R} \quad (10.39)$$

where V is the voltage across the coil terminals and R is the total resistance of the coil. This is,

$$R = 2\sigma\pi r_w l_w \quad (10.40)$$

where r_w is the wire radius, σ is the resistivity of the wire, and the length of the wire is

$$l_w = 2\pi r_b N \quad (10.41)$$

where N is the total number of turns and r_b is the radius of the bar. The wire radius is only available as discrete American Wire Gauge (AWG) values. The dipole moment is

$$M = \pi r_b^2 N \frac{V}{R} \left(1 + \frac{\mu_r - 1}{1 + (\mu_r - 1)N_d} \right) \quad (10.42)$$

where μ_r is the relative permeability and the demagnetizing factor N_d is

$$N_d = \frac{4(\log \gamma - 1)}{\gamma^2 - 4 \log \gamma} \quad (10.43)$$

where

$$\gamma = \frac{l_b}{r_b} \quad (10.44)$$

If we specify power and voltage, we determine R

$$P = IV \quad (10.45)$$

$$V = IR \quad (10.46)$$

$$R = \frac{V^2}{P} \quad (10.47)$$

This leads to the number of turns

$$N = \frac{R}{2\pi r_b \sigma \pi r_w^2} \quad (10.48)$$

where the denominator is the resistance per turn. The number of layers of wire is

$$N_L = \frac{2Nr_w}{l_b} \quad (10.49)$$

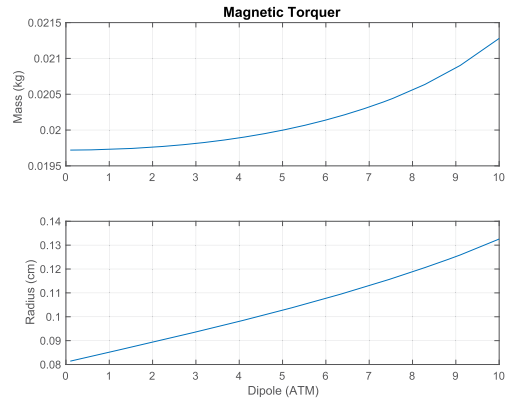
The mass of the torquer is

$$m = \rho_b \pi r_b^2 l_b + 2\rho_w N \pi r_w^2 \pi r_b \quad (10.50)$$

where ρ_b is the density of the bar and ρ_w is the density of the wire. Example 10.7 shows how the mass, dipole, and radius of a torquer varies with terminal voltage. The radius increases by steps because the number of winding layers is an integer. Higher voltages produce more torque per unit mass. This torquer uses Hiperco-A, which has a relative permeability of 15 000. Relative permeabilities above 2000 do not change the mass.

```

1 d.aWG      = 32;
2 d.power    = 4;
3 d.v        = 12;
4 d.dipole    = 4;
5 d.densityConductor = 8960;
6 d.conductivity = 59.6e6;
7 d.densitySteel = 7650;
8 d.muR      = 2000;
9 d.l        = 0.22;
10 dipole     = logspace(-1,1);
11 n          = length(dipole);
12 mass       = zeros(1,n);
13 radius     = zeros(1,n);
14 for k = 1:n
15     d.dipole = dipole(k);
16     d       = MagneticTorquerDesign(d);
17     mass(k)  = d.m;
18     radius(k) = d.rTorquer;
19 end
20 yL = {'Mass(kg)', 'Radius(cm)'};
21
22 Plot2D(dipole,[mass;radius*100],'Dipole(ATM)',yL,
        'MagneticTorquer');
```



Example 10.7: Magnetic-torquer mass and radius vs. voltage.

10.8. Solenoids

10.8.1 Introduction

Solenoids are electromechanical devices that are used in many types of spacecraft systems. This section derives the equations of motion for a dual-coil solenoid. Besides giving models of solenoids, these equations are similar to those used to model rotary and linear motors.

10.8.2 Derivation of the equations of motion for a dual-coil solenoid

The solenoid is illustrated in Fig. 10.9. The first step is to find the magnetic fields that

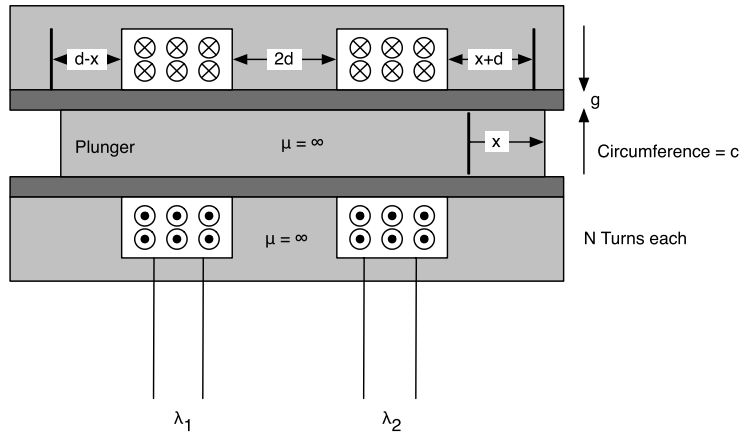


Figure 10.9 Dual-coil solenoid.

are significant only in the sleeve where the permeability is that of free space. In both the plunger and the enclosure the permeability is assumed to be infinite; hence the fields are nearly zero. The magnetic fields are found from the equation

$$-\oint_C H \cdot dl = \int_S J \cdot nda \quad (10.51)$$

This equation means that if you integrate the magnetic fields on a closed path around a surface S , the integral will equal the sum of all of the currents that flow normal to the surface and are within the path. In the above drawing, there are three possible paths. One surrounds just coil 1, one surrounds just coil 2, and one surrounds both coils. There are also three regions in which the magnetic field can be found. One is to the left of coil 1, one is between coil 1 and 2 and the third is to the right of coil 3. Only two paths are required to relate the fields, see Fig. 10.10. The two integrals are

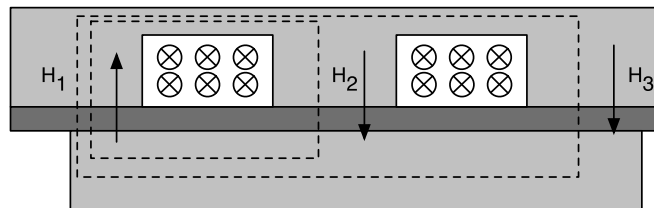


Figure 10.10 Contours for integrating the magnetic field.

$$gH_1 + gH_2 = Ni_1 \quad (10.52)$$

$$gH_1 + gH_3 = N(i_1 + i_2) \quad (10.53)$$

A third relationship is needed to solve for the fields. The surface integral of the magnetic flux over the region in which there are no sources must be zero, or

$$-\oint_S B \cdot nda = 0 \quad (10.54)$$

This surface integral is over a volume, unlike the one in Eq. (14.6). This results in

$$\mu_o c((d-x)H_1 - 2dH_2 - (d+x)H_3) = 0 \quad (10.55)$$

Substituting for H_2 and H_3 gives

$$H_1 = \frac{N}{4g} (2i_1 + (1+x/d)(i_1 + i_2)) \quad (10.56)$$

$$H_2 = \frac{N}{4g} (2i_2 + (1-x/d)(i_1 + i_2)) \quad (10.57)$$

To find the flux linked by each of the coils, it is necessary to find the flux through the volume within each of the coils. For each coil this is

$$\phi_1 = \mu_o cd(1-x/d)H_1 \quad \phi_2 = \mu_o cd(1+x/d)H_3 \quad (10.58)$$

Each flux is linked N times by each coil, so that total flux linked becomes

$$\lambda_1 = \frac{\mu_o N^2 cd}{4g} \left(\left(3 - 2\frac{x}{d} - \frac{x^2}{d^2} \right) i_1 + \left(1 - \frac{x^2}{d^2} \right) i_2 \right) \quad (10.59)$$

$$\lambda_2 = \frac{\mu_o N^2 cd}{4g} \left(\left(1 - \frac{x^2}{d^2} \right) i_1 + \left(1 + 2\frac{x}{d} - \frac{x^2}{d^2} \right) i_2 \right) \quad (10.60)$$

The inductances are

$$L_{11} = L_o \left(3 - 2\frac{x}{d} - \frac{x^2}{d^2} \right) \quad (10.61)$$

$$L_{22} = L_o \left(3 + 2\frac{x}{d} - \frac{x^2}{d^2} \right) \quad (10.62)$$

$$L_{12} = L_o \left(1 - \frac{x^2}{d^2} \right) \quad (10.63)$$

$$L_o = \frac{\mu_o N^2 cd}{4g} \quad (10.64)$$

The terminal voltages are

$$v = iR + \frac{d\lambda}{dt} \quad (10.65)$$

or

$$v_1 = i_1 R + L_{11} \frac{di_1}{dt} + L_{12} \frac{di_2}{dt} - \frac{2L_o}{d} \left(\left(1 + \frac{x}{d}\right) i_1 + \left(\frac{x}{d}\right) i_2 \right) \frac{dx}{dt} \quad (10.66)$$

$$v_2 = i_2 R + L_{121} \frac{di_1}{dt} + L_{22} \frac{di_2}{dt} - \frac{2L_o}{d} \left(\left(\frac{x}{d}\right) i_1 + \left(\frac{x}{d} - 1\right) i_2 \right) \frac{dx}{dt} \quad (10.67)$$

The electrical force is

$$f_e = \frac{\partial W'_m}{\partial x} \quad (10.68)$$

where W'_m is the coenergy. For this example,

$$W'_m = \int_0^{i_1} \lambda_1 di_1 + \int_0^{i_2} \lambda_2 di_2 \quad (10.69)$$

this becomes

$$W'_m = \frac{1}{2} L_{11} i_1^2 + 2L_{12} i_1 i_2 + \frac{1}{2} L_{22} i_2^2 \quad (10.70)$$

and the force on the plunger is

$$f_e = \frac{1}{2} \frac{\partial L_{11}}{\partial x} i_1^2 + 2L_{12} \frac{\partial L_{12}}{\partial x} i_1 i_2 + \frac{1}{2} \frac{\partial L_{22}}{\partial x} i_2^2 \quad (10.71)$$

or

$$f_e = \frac{L_o}{d} \left(\left(1 - \frac{x}{d}\right) i_2^2 - 4 \left(\frac{x}{d}\right) i_1 i_2 - \left(1 + \frac{x}{d}\right) i_1^2 \right) \quad (10.72)$$

$$f_e = \frac{L_o}{d} \left(i_2^2 - i_1^2 - \frac{x}{d} (i_2^2 + i_1^2 + 4i_1 i_2) \right) \quad (10.73)$$

The equation of motion for the mechanical part of the system is

$$M \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + kx + b \frac{dx/dt}{|dx/dt|} = \frac{L_o}{d} \left(i_2^2 - i_1^2 - \frac{x}{d} (i_2^2 + i_1^2 + 4i_1 i_2) \right) \quad (10.74)$$

The first term is the inertial acceleration, the second is viscous damping, the third is linear spring stiffness, and the fourth is Coulomb friction. If the currents are balanced and constant the last term in the coupling term provides stiffness to the system. The mechanical equations are linear in x and the electrical equations are linear in i , and the coupled equations are nonlinear in both x and i .

10.8.3 Derivation of the equations of motion for a single-coil solenoid

The solenoid is illustrated in Fig. 10.11. The first step is to find the H fields. The H

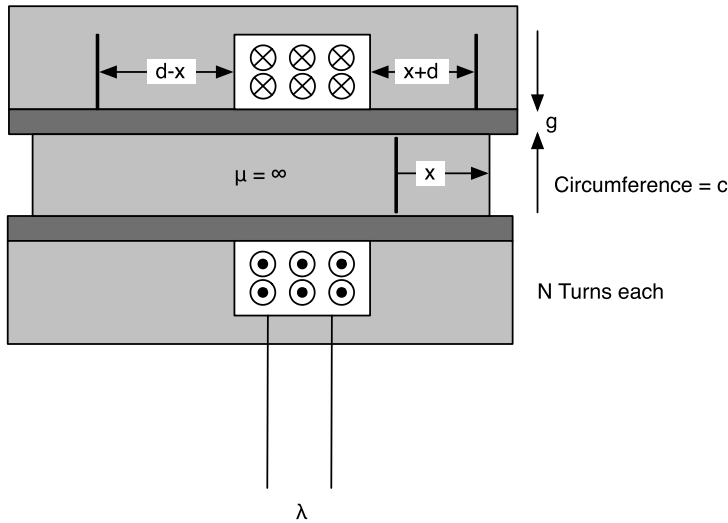


Figure 10.11 Single-coil solenoid.

fields are significant only in the sleeve where the permeability is that of free space. In both the plunger and the enclosure, the permeability is assumed to be infinite; hence the fields are nearly zero. The magnetic fields are found from the equation

$$\oint_C H \cdot dl = \int_S J \cdot nda \quad (10.75)$$

This equation means that if you integrate the magnetic fields on a closed path around a surface S , the integral will equal the sum of all of the currents that flow normal to the surface and are within the path. In the above drawing, there are three possible paths. One surrounds just coil 1, one surrounds just coil 2, and one surrounds both coils. There are also three regions in which the magnetic field can be found. One is to the left of coil 1, one is between coil 1 and 2, and the third is to the right of coil 3. Only two paths are required to relate the fields, see Fig. 10.12. The two integrals are

$$gH_1 + gH_2 = Ni \quad (10.76)$$

A third relationship is needed to solve for the fields. The surface integral of the magnetic flux over a region in which there are no sources must be zero, or

$$\oint_S B \cdot nda = 0 \quad (10.77)$$

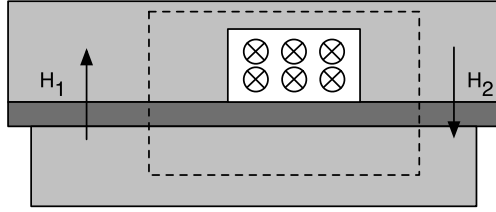


Figure 10.12 Contours for integrating the magnetic field.

This surface integral is over a volume. This results in

$$\mu_0 c((d-x)H_1 - (d+x)H_2) = 0 \quad (10.78)$$

Substituting for H_2 gives

$$H_2 = \frac{N}{g} \left(1 - \frac{x}{d}\right) i \quad (10.79)$$

To find the flux linked by the coil, it is necessary to find the flux through the volume within the coil. This is

$$\phi = \mu_0 c d \left(1 + \frac{x}{d}\right) H_2 \quad (10.80)$$

Each flux is linked N times by the coil so the total flux linked becomes

$$\lambda = \frac{\mu_0 c d N^2}{g} \left(1 - \frac{x^2}{d^2}\right) i \quad (10.81)$$

The inductances are

$$L = L_0 \left(1 - \frac{x^2}{d^2}\right) \quad (10.82)$$

$$L_0 = \frac{\mu_0 c d N^2}{g} \quad (10.83)$$

The terminal voltage is

$$v = iR + \frac{d\lambda}{dt} \quad (10.84)$$

or

$$v = iR + L \frac{di}{dt} - \frac{2L_0 x i}{d^2} \frac{d\lambda}{dt} \quad (10.85)$$

The electrical force is

$$f_e = \frac{\partial W'_m}{\partial x}. \quad (10.86)$$

For this example,

$$\partial W'_m = \int_0^i \lambda i. \quad (10.87)$$

this becomes

$$\partial W'_m = \frac{1}{2} Li^2 \quad (10.88)$$

and the force on the plunger is

$$f_e = \frac{1}{2} \frac{\partial L}{\partial x} i^2 \quad (10.89)$$

or

$$f_e = -\frac{L_o x i^2}{d^2} \quad (10.90)$$

The equations of motion for the system are

$$M \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + kx + b \frac{dx/dt}{|dx/dt|} = -\frac{L_o x i^2}{d^2} \nu = iR + L \frac{di}{dt} - \frac{2L_o x i}{d^2} dx/dt \quad (10.91)$$

The first term is inertial acceleration, the second is linear spring stiffness, and the third is Coulomb friction. When the current is off and motion has damped the damper will sit at $-x_0$. When the current is added, the equation of motion will be

$$M \frac{d^2 x}{dt^2} = \frac{L_o x i^2}{d^2} \quad (10.92)$$

which will push the solenoid in the $+x$ direction. This setup will work well as a valve driver, see Fig. 10.13.

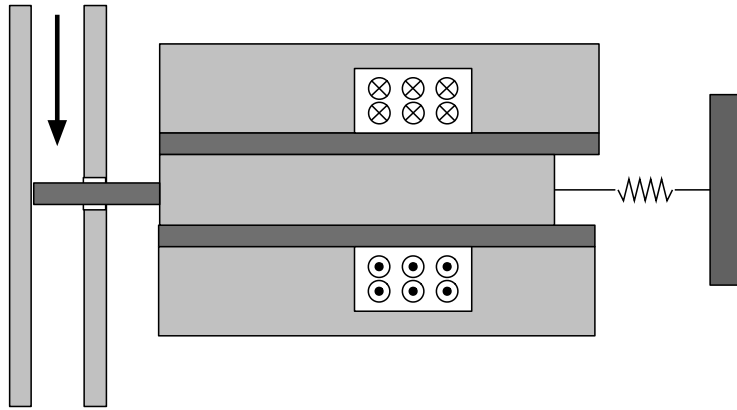


Figure 10.13 Valve driver.

10.9. Stepping motor

Stepping motors can be of many different types including variable reluctance, permanent magnet, hybrid (a combination of variable reluctance and permanent magnet), and claw-tooth permanent magnet, among others. A permanent-magnet motor was chosen as an example to demonstrate how the dynamics of the motor may be handled in models.

A two-phase, permanent-magnet stepping motor consists of a slotted stator with two phases and a permanent-magnet rotor. By energizing the phases of the stator the rotor can be caused to move one step.

The mathematical model of the stepping motor in ab (phase) coordinates is

$$\begin{aligned}
 \frac{di_a}{dt} &= \frac{1}{L} (u_a - Ri_a + K_m \omega s) \\
 \frac{di_b}{dt} &= \frac{1}{L} (u_b - Ri_b - K_m \omega c) \\
 \frac{d\omega}{dt} &= \frac{K_m}{J} \left(i_b c - i_a s - \frac{B}{K_m} \omega \right) \\
 \frac{d\theta}{dt} &= \omega \\
 c &= \cos(N_r \theta) \\
 s &= \sin(N_r \theta)
 \end{aligned} \tag{10.93}$$

The first two equations are the current equations for phases a and b . R is the resistance of the phase winding, K_m is the motor-torque constant, N_r is the number of rotor teeth, J is the rotor inertia, B is the viscous damping coefficient, u_a and u_b are the phase voltages, and i_a and i_b are the phase currents.

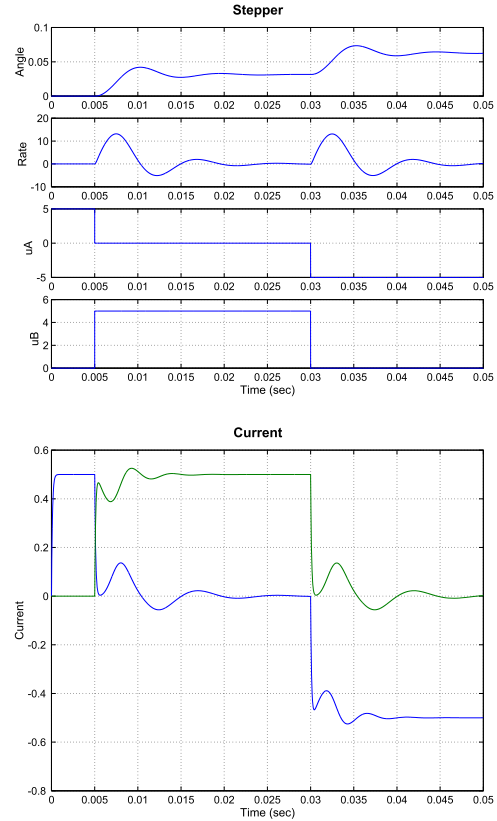
The results from a stepping-motor simulation are shown in Example 10.8. Note that the period of the simulation is only 50 ms. In that time the motor steps twice. Stepping is accomplished by setting the phase voltages to ± 5 V.

The stepping-motor response looks like a moderately damped second-order system. As can be seen from the results, simulating the stepping motor would require 1-ms integration time steps. The dynamics of the stepping motor are far outside the bandwidth of the controller and therefore the approach will be to ignore the transients and assume that when the stepping motor is stepping the mass is moving at the desired velocity all the time. The model is then derived assuming that the mass velocity is constrained. When the mass velocity changes, the core-body rates are changed instantaneously so that angular momentum is conserved.

```

1 tSim      = 0.05;
2 dT        = 0.000005;
3 nSim      = tSim/dT;
4 dS.l      = 0.0011; % Inductance
5 dS.nR     = 50;      % Number of teeth
6 dS.r      = 10;      % Resistance
7 dS.kM     = 0.113;   % Motor torque
8 dS.kD     = 0;       % Detente torque
9 dS.tM     = 0.0;      % Mechanical shaft torques
10 dS.coulomb = 0.0;
11 dS.inertia = 5.7e-6;
12 dS.damping = 0.001;
13 dS.mode   = 'ab';
14
15 xPlot     = zeros(6,nSim);
16 tPlot     = zeros(1,nSim);
17
18 x         = [0;0;0;0];
19 t         = 0;
20 dS.uA     = 5;
21 dS.uB     = 0;
22
23 for k = 1:nSim
24     if ( t >= 0.005 && t <= 0.03 )
25         dS.uA = 0;
26         dS.uB = 5;
27     elseif ( t > 0.03 )
28         dS.uA = -5;
29         dS.uB = 0;
30     end
31
32     xPlot(:,k) = [x;dS.uA;dS.uB];
33     tPlot(k)   = t;
34     x          = RK4('StepSim',x,dT,t,dS);
35     t          = t + dT;
36 end
37 g = [1 2 5 6];
38 Plot2D( tPlot, xPlot(g,:), 'Time(sec)', ...
39 {'Angle' 'Rate', 'uA' 'uB'}, 'Stepper')
40 Plot2D( tPlot, xPlot(3:4,:), 'Time(sec)', '
    Current', 'Current')

```



Example 10.8: Stepping motor.

10.10. Dampers

Dampers are used to dissipate energy in a spacecraft. Fuel slosh can act as a damper. Mechanical joints also provide damping. For example, deployable solar panels have numerous joints and they all act as dampers.

Appendix L gives the theory about magnetoresistive dampers, a particularly simple type of damper. Ball-in-tube dampers are also used on spacecraft.

References

- [1] C.C. de Wit, H. Olsson, K.J. Astrom, P. Lischinsky, A new model for control of systems with friction, IEEE Transactions on Automatic Control 40 (3) (March 1995) 419–425.
- [2] M. Hasha, Passive isolation/damping system for the Hubble Space Telescope reaction wheels, in: The 21st Aerospace Mechanisms Symposium, NASA-Lyndon B. Johnson Space Center, May 1987.